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Mid-Term answer sheet

Q1) Overfitting is when the model is too complex and gets high accuracy on training data but low accuracy on testing data.
To reduce it we can use regularization. Another method is to use more data.

Q2) Self attention lets each word look at the other words in the sentence. We create query key and value. The query is multiplied with the keys to get scores and these scores tell how important each word is. Then softmax is applied to make them probabilities. The values are then multiplied by these probabilities and added up to get the output.
We divide by root of dk to make the computation faster.

Q3) Precision = TP / (TP + FN) = 40/60 = 0.67
Recall = TP / (TP + FP) = 40/50 = 0.8
F1 = 2 x 0.67 x 0.8 / (0.67 + 0.8) = 0.729
Accuracy = 40+30 / 100 = 0.7

(continued)

Q4) GANs have a generator and a discriminator which compete with each other. Diffusion models add noise to images and then learn to remove it.
GANs are harder to train than diffusion models. GANs have mode collapse problem.
Diffusion models are better quality but slower.

Q5) In RAG we retrieve relevant documents from a database and give them to the LLM so it can answer questions with up to date information. The documents are stored as embeddings.
One problem is hallucination. We can solve it by using a bigger model.
Another problem is the retrieval can be slow when there are many documents.

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Mid-Term answer sheet

Q1. Overfitting happens when a model learns the training examples too well, including random noise, so it shows very low training error but high error on validation/test data.
1) L1/L2 regularization - adds a penalty for large weights so the model becomes simpler.
2) Data augmentation - creating more varied training examples (flips, crops) so the model can't just memorise them.

Q2. Each token gets Q, K and V vectors by multiplying its embedding with weight matrices that are learned in training. Score = Q.K for every pair of tokens, then softmax converts the scores into weights that sum to one, and the output is the weighted sum of the V vectors. We scale by sqrt(dk) to keep the numbers small.

Q3. P = 40/(40+10) = 0.8
R = 40/(40+20) = 0.67
F1 = 2*0.8*0.67/(0.8+0.67) = 0.73
Accuracy = (40+30)/100 = 0.7

(continued)

Q4. GAN: a generator and a discriminator play a minimax game. Diffusion: the model learns to denoise images that were corrupted with Gaussian noise over many timesteps.
GAN training is unstable and can have mode collapse; diffusion training is stable and the samples are more diverse.
GANs are fast (a single forward pass), diffusion is slow (hundreds of denoising steps).

Q5. RAG: chunk the documents, embed each chunk with an embedding model and store the vectors in a vector DB like FAISS. At query time embed the question, retrieve the top-k similar chunks and put them in the prompt for the LLM.
Failure: chunks too large or too small, so the right information is split or lost -> tune the chunk size and use overlap between chunks.
Another issue is cost.